

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

Section 1: Physics

I. General Physics & Measurement

1. SI Base Units: Length (m), Mass (kg), Time (s), Current (A), Temperature (K), Amount of substance (mol).
2. Scalar Quantities: Have magnitude only (e.g., speed, distance, mass, energy, density).
3. Vector Quantities: Have both magnitude and direction (e.g., velocity, displacement, force, acceleration, momentum).
4. Precision vs. Accuracy: Precision is repeatability; accuracy is closeness to the true value.
5. Micrometer Screw Gauge: Measures small lengths up to 25 mm with precision to 0.01 mm.
6. Vernier Calipers: Measure internal/external diameters and depth with precision to 0.01 cm.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

II. Mechanics & Motion

7. Speed: Distance traveled per unit time (Speed = Distance / Time).
8. Velocity: Rate of change of displacement (Velocity = Displacement / Time).
9. Acceleration: Rate of change of velocity ($a = (v - u) / t$).
10. Gradient of Distance-Time Graph: Represents speed.
11. Gradient of Velocity-Time Graph: Represents acceleration.
12. Area under Velocity-Time Graph: Represents total displacement/distance.
13. Mass: Amount of matter in a body; scalar; constant everywhere (measured in kg).
14. Weight: Force of gravity acting on a mass ($W = mg$); vector; varies with gravitational field strength (g).

15. Density: Mass per unit volume (Density = Mass / Volume).
16. Center of Gravity: The single point through which the entire weight of an object appears to act.
17. Stability: Higher stability is achieved with a lower center of gravity and a wider base area.
18. Newton's First Law (Inertia): An object remains at rest or constant velocity unless acted upon by a resultant force.
19. Newton's Second Law: Resultant force equals mass times acceleration ($F = ma$).
20. Newton's Third Law: For every action, there is an equal and opposite reaction force.
21. Friction: Force opposing motion between two surfaces in contact; generates heat.
22. Terminal Velocity: Reached when weight equals air resistance; acceleration becomes zero.
23. Moment of a Force: Turning effect of a force about a pivot (Moment = Force x perpendicular distance).
24. Principle of Moments: For equilibrium, Sum of Clockwise Moments = Sum of Anticlockwise Moments.
25. Conditions for Equilibrium: Zero resultant force and zero resultant turning effect.
26. Hooke's Law: Extension is directly proportional to load applied, provided the limit of proportionality is not exceeded ($F = kx$).
27. Pressure: Force per unit area ($P = F / A$). Measured in Pascals (Pa).
28. Liquid Pressure: Increases with depth and fluid density ($P = h * \text{density} * g$).
29. Barometer: Used to measure atmospheric pressure (standard atmospheric pressure = 760 mmHg = 101,300 Pa).
30. Manometer: Used to measure gas pressure differences.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For

questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

III. Energy, Work & Power

- 31. Work Done: Product of force and distance moved in the direction of the force ($W = F \cdot d$). Unit: Joule (J).
 - 32. Kinetic Energy: Energy due to motion ($E_k = 0.5 \cdot m \cdot v^2$).
 - 33. Gravitational Potential Energy: Energy stored due to height ($E_p = m \cdot g \cdot h$).
 - 34. Principle of Conservation of Energy: Energy cannot be created or destroyed, only transformed from one form to another.
 - 35. Efficiency: Efficiency = (Useful Energy Output / Total Energy Input) * 100%.
 - 36. Power: Rate of doing work or transferring energy ($P = \text{Work} / \text{time} = \text{Energy} / \text{time}$). Unit: Watt (W).
 - 37. Renewable Energy Sources: Solar, wind, hydroelectric, tidal, geothermal, biomass.
 - 38. Non-Renewable Energy Sources: Fossil fuels (coal, oil, gas), nuclear fuels.
- PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

IV. Thermal Physics

- 39. Kinetic Particle Theory: Solids have fixed volume/shape; liquids have fixed volume/variable shape; gases fill containers.
- 40. Brownian Motion: Random movement of microscopic particles suspended in a fluid due to collisions with fast-moving molecules.
- 41. Thermal Expansion: Solids expand least, gases expand most when heated.
- 42. Conduction: Heat transfer through solid particle vibrations and free electron diffusion (in metals).
- 43. Convection: Heat transfer in fluids driven by density differences (hot fluid rises, cooler fluid sinks).
- 44. Radiation: Infrared heat transfer via electromagnetic waves; requires no medium.
- 45. Emitters and Absorbers: Dull, black surfaces are the best absorbers and emitters; shiny, white surfaces reflect radiation.
- 46. Specific Heat Capacity: Thermal energy needed to raise 1 kg of substance by 1 degree Celsius ($Q = m \cdot c \cdot \Delta T$).

47. Latent Heat of Fusion: Energy required to change unit mass from solid to liquid without temperature change ($Q = m \cdot L_f$).

48. Latent Heat of Vaporization: Energy required to change unit mass from liquid to gas without temperature change ($Q = m \cdot L_v$).

49. Melting & Boiling: Occur at constant temperatures.

50. Evaporation: Surface phenomenon occurring at any temperature below boiling; causes cooling.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

V. Waves, Light & Sound

51. Transverse Waves: Oscillations are perpendicular to wave direction (e.g., light, water waves).

52. Longitudinal Waves: Oscillations are parallel to wave direction (e.g., sound waves).

53. Wave Equation: Wave speed equals frequency times wavelength ($v = f \cdot \text{wavelength}$).

54. Frequency (f): Number of complete oscillations per second ($f = 1 / T$). Measured in Hertz (Hz).

55. Refraction: Change in wave speed and direction when entering a medium of different optical density.

56. Snell's Law: Refractive index $n = \sin(i) / \sin(r) = \text{speed in vacuum} / \text{speed in medium}$.

57. Critical Angle (c): Angle of incidence in dense medium yielding an angle of refraction of 90 degrees ($\sin(c) = 1 / n$).

58. Total Internal Reflection: Occurs when light travels from denser to rarer medium and angle of incidence exceeds critical angle.

59. Convex (Converging) Lens: Focuses parallel rays to a real focal point; real or virtual images depending on object distance.

60. Electromagnetic Spectrum Order (Decreasing Wavelength): Radio -> Microwave -> Infrared -> Visible -> UV -> X-rays -> Gamma rays.

61. EM Wave Speed: All EM waves travel at **$3 \times 10^8 \text{ m/s}$ in vacuum**.

62. Sound Waves: Longitudinal mechanical waves requiring a medium

(cannot travel through vacuum).

63. Speed of Sound: Highest in solids, lower in liquids, lowest in gases (approx. 330 m/s in air).

64. Pitch & Loudness: Pitch depends on frequency; loudness depends on amplitude.

65. Human Hearing Range: 20 Hz to 20,000 Hz.

66. Electric Current: Rate of flow of electric charge ($I = Q / t$). Unit: Ampere (A).

67. Electromotive Force (e.m.f.): Work done by source in driving unit charge around a complete circuit.

68. Potential Difference (p.d.): Work done to drive unit charge across a component ($V = W / Q$). Unit: Volt (V).

69. Ohm's Law: Current is directly proportional to p.d. across a conductor at constant temperature ($V = I * R$).

70. Resistance of a Wire: $R = \text{resistivity} * (\text{Length} / \text{Area})$ (proportional to length, inversely proportional to cross-sectional area).

71. Series Circuits: Current same throughout; total voltage sums up ($V_T = V_1 + V_2$); $R_T = R_1 + R_2$.

72. Parallel Circuits: Voltage same across branches; current splits; $1/R_T = 1/R_1 + 1/R_2$.

73. Electrical Power Equations: $P = V * I = I^2 * R = V^2 / R$.

74. Electrical Energy Equation: $E = P * t = V * I * t$.

75. Live, Neutral, Earth Wires: Live (brown - carries high voltage), Neutral (blue - completes circuit), Earth (yellow/green - safety path).

76. Fuses & Circuit Breakers: Installed in live wire to protect against current overload.

77. Magnetic Field Lines: Point from North Pole to South Pole.

78. Soft vs. Hard Magnetic Materials: Soft iron magnetizes/demagnetizes easily (electromagnets); steel retains magnetism (permanent magnets).

79. Right-Hand Grip Rule: Determines direction of magnetic field around a current-carrying wire.

80. Fleming's Left-Hand Rule: Used for motors (Thumb = Motion, First finger = Field, Second finger = Current).

81. Electromagnetic Induction: e.m.f. is

induced when magnetic flux linkages change (Faraday's Law).

82. Lenz's Law: Induced current flows in a direction that opposes the change causing it.

83. Transformers: Step-up ($N_s > N_p$) increases voltage; Step-down ($N_s < N_p$) decreases voltage ($V_p / V_s = N_p / N_s$).

84. Radioactive Radiation Types: Alpha (Helium nucleus), Beta (high-speed electron), Gamma (EM wave).

85. Half-Life: Time taken for half the radioactive nuclei in a sample to decay.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

Section 2: Chemistry

I. Principles of Chemistry & Matter

86. Atom Structure: Protons (+1) and neutrons (0) in nucleus; electrons (-1) orbiting in shells.

87. Atomic Number (Z): Number of protons in an atom's nucleus.

88. Mass/Nucleon Number (A): Total number of protons and neutrons ($A = Z + N$).

89. Isotopes: Atoms of the same element with same atomic number but different mass numbers (different neutron count).

90. Elements: Substances made of only one type of atom.

91. Compounds: Two or more different elements chemically combined in fixed proportions.

92. Mixtures: Substances physically combined without fixed proportions; easily separated physically.

93. Filtration: Separates an insoluble solid from a liquid.

94. Crystallization: Obtains a dissolved solid from a saturated solution.

95. Simple Distillation: Separates a solvent from a solution with different boiling points.

96. Fractional Distillation: Separates miscible liquids with close boiling points (e.g., crude oil, ethanol/water).

97. Paper Chromatography: Separates soluble substances based on differential solubility in a solvent.

98. Rf Value: $R_f = \text{Distance moved by solute} / \text{Distance moved by solvent front}$.

II. Chemical Bonding & Structure

99. Ionic Bonding: Transfer of electrons from metal to non-metal forming cations and anions.
100. Giant Ionic Lattice: High melting points, soluble in water, conducts electricity when molten or aqueous (not solid).
101. Covalent Bonding: Sharing of electron pairs between non-metal atoms.
102. Simple Molecular Structures: Low melting/boiling points due to weak intermolecular forces; non-conductors.
103. Giant Covalent Structures: Diamond (hard, non-conductor), Graphite (layers, soft, conducts electricity due to delocalized electrons), SiO₂.
104. Metallic Bonding: Electrostatic attraction between positive metal ions and delocalized sea of electrons.
105. Properties of Metals: Good conductors of heat/electricity, malleable, ductile.
106. Alloys: Mixtures of a metal with other elements; harder than pure metals because different atomic sizes disrupt regular layers.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

III. Stoichiometry & Chemical Calculations

107. Mole Concept: 1 mole = 6.02×10^{23} particles (Avogadro's constant).
108. Moles from Mass: Moles = Mass (g) / Molar Mass (g/mol).
109. Moles of Gas: Moles = Volume of gas (dm³) / 24 dm³ at standard room temperature and pressure (s.t.p.).
110. Moles from Solution: Moles = Concentration (mol/dm³) * Volume (dm³).
111. Empirical Formula: Simplest whole-number ratio of atoms of each element in a compound.
112. Molecular Formula: Actual number of atoms of each element in a molecule.
113. Percentage Yield: (Actual Yield / Theoretical Yield) * 100%.
114. Percentage Purity: (Mass of Pure

Substance / Total Mass of Impure Sample) * 100%.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

IV. Acids, Bases & Salts

115. Acids: Produce H⁺ ions in aqueous solution; proton donors; pH < 7.
116. Bases: Proton acceptors (oxides and hydroxides of metals).
117. Alkalis: Water-soluble bases; produce OH⁻ ions in aqueous solution; pH > 7.
118. Neutralization: H⁺ (aq) + OH⁻ (aq) → H₂O (l).
119. Acid Reactions:
- * Acid + Metal → Salt + Hydrogen Gas
 - * Acid + Base → Salt + Water
 - * Acid + Carbonate → Salt + Water + Carbon Dioxide
120. Solubility Rules: All Sodium, Potassium, Ammonium, and Nitrate salts are soluble (SPAN).
121. Sulfate Solubility: All soluble except Lead, Barium, and Calcium sulfates.
122. Chloride Solubility: All soluble except Lead and Silver chlorides.
123. Precipitation: Method used to prepare insoluble salts by mixing two soluble salt solutions.
124. Oxides: Basic (metals), Acidic (non-metals), Amphoteric (Al₂O₃, ZnO, PbO), Neutral (CO, NO, H₂O).

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

V. Redox & Electrochemistry

125. Oxidation: Gain of oxygen, loss of hydrogen, loss of electrons, or increase in oxidation state.
126. Reduction: Loss of oxygen, gain of hydrogen, gain of electrons, or decrease in oxidation state.
127. Oxidizing Agent: Gets reduced; accepts electrons (e.g., Acidified KMnO₄ turns purple to colorless).
128. Reducing Agent: Gets oxidized; donates electrons (e.g., KI turns colorless to brown).
129. Electrolysis: Breakdown of an ionic compound (molten or aqueous) by

electricity.

130. Anode: Positive electrode (oxidation occurs here).

131. Cathode: Negative electrode (reduction occurs here).

132. Electrolysis of Molten NaCl: Sodium at cathode; Chlorine gas at anode.

133. Electrolysis of Aqueous CuSO₄ with Carbon Electrodes: Copper deposited at cathode; O₂ gas formed at anode.

134. Electroplating: Object to be coated placed at cathode; coating metal placed at anode.

VI. Energetics & Reaction Kinetics

135. Exothermic Reactions: Release heat energy to surroundings; Delta H is negative (e.g., combustion, neutralization).

136. Endothermic Reactions: Absorb heat energy from surroundings; Delta H is positive (e.g., thermal decomposition, photosynthesis).

137. Activation Energy (E_a): Minimum energy colliding particles need to react.

138. Bond Making/Breaking: Bond breaking is endothermic; bond making is exothermic.

139. Factors Increasing Reaction Rate: Increasing concentration, pressure, surface area, temperature, or using a catalyst.

140. Catalysts: Speed up chemical reactions by lowering activation energy without being consumed.

141. Reversible Reactions: Can proceed forward and backward; reach dynamic equilibrium in closed systems.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/v4sk57>

VII. Periodic Table, Metals & Environment

142. Group 1 (Alkali Metals): Reactivity increases down the group; soft, low density.

143. Group 7 (Halogens): Reactivity decreases down the group; melting points increase down the group.

144. Group 0/8 (Noble Gases): Unreactive/inert due to full outer electron shells.

145. Transition Metals: Form colored compounds, variable oxidation states, high densities, useful as catalysts.

146. Reactivity Series (Highest to Lowest):

K, Na, Ca, Mg, Al, C, Zn, Fe, Pb, H, Cu, Ag, Au.

147. Extraction of Metals: Metals above Carbon extracted by electrolysis; below Carbon extracted by reduction with Carbon/CO.

148. Blast Furnace Operations: Hematite (Fe₂O₃) reduced by CO to Iron; Limestone removes silica impurities as slag (CaSiO₃).

149. Rusting Conditions: Requires both oxygen and water.

150. Rust Prevention: Galvanizing (coating with zinc), barrier methods (paint/grease), sacrificial protection.

151. Air Composition: 78% N₂, 21% O₂, 0.9% Ar, 0.04% CO₂.

152. Air Pollutants: Carbon Monoxide (toxic), Sulfur Dioxide (acid rain), Nitrogen Oxides (acid rain/smog).

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/v4sk57>

VIII. Organic Chemistry

153. Hydrocarbons: Compounds containing hydrogen and carbon atoms only.

154. Alkanes: Saturated hydrocarbons (C_nH_{2n+2} contain single bonds only.

155. Alkenes: Unsaturated hydrocarbons (C_nH_{2n}); contain C=C double bonds.

156. Test for Unsaturation: Aqueous Bromine turns from orange/brown to colorless with alkenes.

157. Fractional Distillation Fractions (Top to Bottom): Refinery gas, Gasoline, Naphtha, Kerosene, Diesel, Fuel oil, Lubricating oil, Bitumen.

158. Cracking: Thermal breakdown of long-chain alkanes into shorter alkanes and useful alkenes.

159. Alcohols: Contain -OH functional group (C_nH_{2n+1}OH). Manufactured via fermentation or hydration of ethene.

160. Carboxylic Acids: Contain -COOH functional group (C_nH_{2n+1}COOH). Weak organic acids.

161. Esters: Formed by reacting an alcohol with a carboxylic acid in presence of acid catalyst; have sweet/fruity smells.

162. Polymers: Large molecules made of repeating monomer units (Addition: polyethene; Condensation: nylon, terylene).

Section 3: Biology

I. Cell Structure & Organization

- 163. Cell Membrane: Controls entry and exit of substances; selectively permeable.
- 164. Cell Wall: Made of cellulose in plants; provides structural support and prevents bursting.
- 165. Nucleus: Contains genetic material (DNA); controls cellular activities.
- 166. Cytoplasm: Site where most metabolic reactions occur.
- 167. Mitochondria: Site of aerobic respiration; produces energy currency (ATP).
- 168. Chloroplasts: Contain chlorophyll; site of photosynthesis in plant cells.
- 169. Ribosomes: Site of protein synthesis.
- 170. Vacuole: Large, central in plant cells (contains cell sap); small and temporary in animal cells.
- 171. Plant vs. Animal Cells: Plants have cell walls, chloroplasts, and large central vacuoles; animal cells do not.
- 172. Specialized Cells: Red blood cells (no nucleus, biconcave for O₂ transport), Root hair cells (large surface area for absorption), Xylem vessels (hollow, lignified for water transport).

II. Transport Mechanisms

- 173. Diffusion: Net movement of particles from higher to lower concentration down a concentration gradient (passive).
- 174. Osmosis: Net movement of water molecules from higher water potential to lower water potential across a partially permeable membrane.
- 175. Turgor Pressure: Internal pressure exerted by water inside a plant cell pushing against the cell wall.
- 176. Plasmolysis: Shrinkage of cytoplasm away from cell wall when plant cell is placed in hypertonic solution.
- 177. Active Transport: Movement of particles against concentration gradient

using cellular energy (ATP) via carrier proteins.

III. Biological Molecules & Enzymes

- 178. Carbohydrates: Made of C, H, O. Monosaccharides (glucose), Disaccharides (sucrose), Polysaccharides (starch, glycogen, cellulose).
- 179. Proteins: Made of C, H, O, N (sometimes S). Composed of long chains of amino acids.
- 180. Lipids (Fats): Made of C, H, O. Composed of 1 glycerol and 3 fatty acid molecules.
- 181. Food Tests:

- * Starch: Iodine solution turns from brown to blue-black.
- * Reducing Sugars: Benedict's reagent turns from blue to green/yellow/red precipitate when heated.
- * Protein: Biuret reagent turns from blue to purple/violet.
- * Fats: Ethanol emulsion test forms a cloudy white emulsion also OSMIC TESTS to give a black ppt.

- 182. Enzymes: Biological and organic catalysts made of protein that speed up metabolic reactions without being consumed.

- 183. Lock and Key Model: Substrate binds specifically to the enzyme's active site.

- 184. Denaturation: High temperatures or extreme pH alter active site shape, preventing substrate binding permanently.

IV. Plant Nutrition & Transport

- 185. Photosynthesis Equation: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (in presence of light and chlorophyll).
- 186. Chlorophyll: Green pigment that traps light energy for photosynthesis.
- 187. Leaf Structure: Upper epidermis -> Palisade mesophyll (packed with chloroplasts) -> Spongy mesophyll (air spaces for gas exchange) -> Stomata/Guard cells.
- 188. Limiting Factors of Photosynthesis:

Light intensity, CO₂ concentration, temperature.

189. Xylem: Transport water and mineral ions unidirectionally from roots to leaves.

190. Phloem: Translocate sucrose and amino acids bidirectionally from sources to sinks.

191. Transpiration: Evaporation of water vapor from aerial parts of plants (mainly via leaf stomata).

192. Factors Increasing Transpiration: High temperature, low humidity, high wind speed, high light intensity.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

V. Human Nutrition & Digestion

193. Balanced Diet: Contains carbohydrates, proteins, lipids, vitamins, minerals, water, and dietary fiber in correct proportions.

194. Vitamin Deficiencies: Vitamin C causes scurvy; Vitamin D causes rickets.

195. Mineral Deficiencies: Calcium causes weak bones/teeth; Iron causes anemia.

196. Peristalsis: Wave-like muscle contractions pushing food along alimentary canal. We also have anti-peristalsis

197. Amylase: Breaks down starch into maltose (salivary glands, pancreas).

198. Protease (e.g., Pepsin, Trypsin): Breaks down proteins into amino acids (pepsin in acidic stomach).

199. Lipase: Breaks down emulsified lipids into fatty acids and glycerol (pancreas).

200. Bile: Produced in liver, stored in gallbladder; neutralizes stomach acid and emulsifies fats (increases surface area).

201. Villi Absorption: Small intestine wall adapted with microvilli, thin walls, and dense capillary networks for absorption.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

VI. Human Respiration & Gas Exchange

202. Aerobic Respiration: $C_6H_{12}O_6 + 6 O_2 \rightarrow 6 CO_2 + 6 H_2O + \text{Energy (ATP)}$.

203. Anaerobic Respiration (Humans): Glucose $\rightarrow$ Lactic Acid + Energy (occurs during heavy exercise).

204. Anaerobic Respiration (Yeast): Glucose $\rightarrow$ Ethanol + Carbon Dioxide + Energy (Fermentation).

205. Oxygen Debt: Amount of oxygen

required after exercise to break down accumulated lactic acid in the liver.

206. Alveoli Adaptations: Large surface area, thin walls (one-cell thick), moist surfaces, rich blood supply for gas exchange.

207. Inhalation: Diaphragm contracts and flattens; external intercostal muscles contract; thorax volume increases, pressure drops.

VII. Human Transport & Immunity

208. Components of Blood: Plasma (liquid carrier), Red blood cells (oxygen transport), White blood cells (immunity), Platelets (blood clotting).

209. Phagocytes vs. Lymphocytes: Phagocytes engulf pathogens (phagocytosis); lymphocytes produce specific antibodies. *NOTE: there is difference between lymphocytes and leucocytes*

210. Double Circulation: Blood passes through heart twice per complete circuit (Pulmonary and Systemic circuits).

211. Heart Structure: Left ventricle has thicker muscular wall than right ventricle to pump blood around entire body.

212. Arteries: Carry blood away from heart under high pressure; thick muscular walls, narrow lumen, no valves.

213. Veins: Carry blood to heart under low pressure; thin walls, wide lumen, contain valves to prevent backflow.

214. Capillaries: Wall one cell thick for rapid substance exchange with body tissues.

215. Coronary Heart Disease: Blockage of coronary arteries supplying heart muscle with oxygenated blood. *Also note that the coronary artery supplies the heart itself.*

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

VIII. Excretion & Homeostasis

216. Excretion: Removal of toxic materials, waste products of metabolism, and excess substances.

217. Kidney Function: Excretes urea, excess salts, and water as urine.

218. Nephron Processes: Ultrafiltration at glomerulus; selective reabsorption of glucose and essential ions in tubules.

219. Homeostasis: Maintenance of a

constant internal environment (e.g., body temperature, blood glucose concentration). NOTE: *there is difference between **hemostasis** and **homeostasis***

220. Negative Feedback: Mechanism that detects deviation from set point and triggers responses to reverse the change.

221. Insulin: Hormone from pancreas that lowers blood glucose by converting glucose to glycogen in the liver.

222. Glucagon: Hormone from pancreas that raises blood glucose by converting stored glycogen back into glucose.

223. Thermoregulation (Cold): Vasoconstriction (narrowing arterioles near skin), shivering, hair erection.

224. Thermoregulation (Hot): Vasodilation (widening arterioles near skin), sweating, flattening hairs.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

IX. Coordination, Response & Reproduction

225. Central Nervous System (CNS): Comprises the brain and spinal cord.

226. Reflex Arc Pathway: Stimulus -> Receptor -> Sensory Neuron -> Relay Neuron -> Motor Neuron -> Effector -> Response.

227. Synapse: Microscopic gap between two neurons; signal transmitted chemically via neurotransmitters.

228. Accommodation (Eye Focus):

- * Distant Objects: Ciliary muscles relax, suspensory ligaments pull tight, lens becomes thin.

- * Near Objects: Ciliary muscles contract, suspensory ligaments slacken, lens becomes fat/convex.

229. Hormones: Chemical substances produced by glands, transported in blood, altering activity of specific target organs.

230. Adrenaline: "Flight or fight" hormone; increases heart rate, breathing rate, and blood glucose.

231. Tropisms in Plants: Phototropism (response to light) and Gravitropism (response to gravity) mediated by Auxin.

232. Asexual Reproduction: Production of genetically identical offspring (clones) from one parent without gamete fusion.

233. Sexual Reproduction: Fusion of haploid male and female gametes to produce a genetically unique diploid zygote.

234. Flower Structure: Anther and Filament (Stamen/Male); Stigma, Style, and Ovary (Carpel/Female).

235. Pollination: Transfer of pollen grains from anther to stigma (by wind or insects).

236. Menstrual Cycle:

- * FSH: Stimulates follicle development.

- * Estrogen: Repairs uterus lining.

- * LH: Triggers ovulation on Day 14.

- * Progesterone: Maintains thick uterus lining.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

X. Genetics, Ecology & Environment

237. Chromosome: Thread-like structure of DNA carrying genetic information in the form of genes.

238. Gene: Length of DNA coding for a specific functional protein.

239. Mitosis: Nuclear division giving rise to 2 genetically identical diploid daughter cells (growth and repair).

240. Meiosis: Reduction division giving rise to 4 genetically different haploid gametes.

241. Alleles: Alternative forms of the same gene (e.g., dominant vs. recessive).

242. Genotype vs. Phenotype: Genotype is genetic makeup (e.g., Bb); phenotype is physical appearance (e.g., brown eyes).

243. Homozygous vs. Heterozygous: Homozygous has identical alleles (BB or bb); heterozygous has different alleles (Bb).

244. Monohybrid Cross: Heterozygous parent cross (Bb x Bb) yields a phenotype ratio of 3:1.

245. Mutation: Random change in gene structure or chromosome number (e.g., Down syndrome = extra chromosome 21).

246. Natural Selection: Survival and reproduction of organisms with advantageous traits passed to future generations.

247. Food Chain: Directs flow of energy from primary Producers -> Primary Consumers -> Secondary Consumers.

248. Energy Loss in Food Chains: Approx.

90% of energy is lost at each trophic level via heat, respiration, and excretion.

249. Carbon Cycle: Process driven by photosynthesis (removes CO_2), respiration, combustion, and decomposition (returns CO_2).

250. Eutrophication: Excessive fertilizer runoff leads to algae blooms, blocked sunlight, aquatic plant death, bacterial growth, and oxygen depletion.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

I. Scientific Method & Experimental Design

251. Independent Variable: The factor intentionally changed or manipulated in an experiment.

252. Dependent Variable: The factor measured or observed to assess the effect of changing the independent variable.

253. Control Variables: Factors kept constant throughout an experiment to ensure a fair test.

254. Control Experiment: A baseline setup where the independent variable is omitted to prove it causes the observed effect.

255. Anomalous Data: Outlier readings that deviate significantly from the general trend; excluded when calculating averages.

256. Parallax Error: Measurement error occurring when a scale is read at an oblique angle rather than perpendicular at eye level.

257. Systematic Error: Consistent error caused by faulty equipment or improper calibration (e.g., zero error).

258. Random Error: Unpredictable variation in measurement caused by environmental changes or human reaction speed.

259. Safety Protocols: Wear safety goggles for heating/chemicals; use water baths for flammable liquids (ethanol); tie back long hair.

260. Meniscus Reading: Read the volume of clear liquids at the bottom of the curved surface (meniscus) at eye level.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

Laboratory Tests & Qualitative Analysis

261. Gas Test - Oxygen (O_2): Relights a glowing splint.

262. Gas Test - Hydrogen (H_2): Extinguishes a burning splint with a characteristic "pop" sound.

263. Gas Test - Carbon Dioxide (CO_2): Turns limewater (calcium hydroxide solution) cloudy/milky.

264. Gas Test - Ammonia (NH_3): Damp red litmus paper turns blue; sharp pungent odor.

265. Gas Test - Chlorine (Cl_2): Bleaches damp litmus paper white.

266. Flame Test - Lithium (Li^+): Crimson red flame.

267. Flame Test - Sodium (Na^+): Persistent yellow flame.

268. Flame Test - Potassium (K^+): Lilac flame.

269. Flame Test - Copper(II) (Cu^{2+}): Blue-green flame.

270. Flame Test - Calcium (Ca^{2+}): Brick red flame.

271. Anion Test - Carbonate (CO_3^{2-}): Add dilute acid; effervescence observed; gas turns limewater cloudy.

272. Anion Test - Chloride (Cl^-): Acidify with dilute HNO_3 , add aqueous AgNO_3 ; forms a white precipitate.

273. Anion Test - Sulfate (SO_4^{2-}): Acidify with dilute HNO_3 , add aqueous $\text{Ba}(\text{NO}_3)_2$; forms a white precipitate.

274. Cation Test - Iron(II) (Fe^{2+}): Add aqueous NaOH ; forms a green precipitate, insoluble in excess.

275. Cation Test - Iron(III) (Fe^{3+}): Add aqueous NaOH ; forms a reddish-brown precipitate, insoluble in excess.

276. Cation Test - Copper(II) (Cu^{2+}): Add aqueous NaOH ; forms a light blue precipitate; soluble in excess aqueous ammonia to form a dark blue solution.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>

III. Some Core Concepts

277. Conservation of Mass: Matter cannot be created or destroyed in chemical or physical processes; total mass of reactants equals total mass of products.

278. Energy Transfer Efficiency: Energy always dissipates into less useful thermal

forms, meaning no machine or biological process operates at 100% efficiency.

279. Surface Area to Volume Ratio: As an organism or particle increases in size, its surface-area-to-volume ratio decreases, altering rates of heat loss, diffusion, and chemical reactivity.

280. Osmoregulation & Turgidity: Water movement across biological and artificial membranes relies on water potential gradients, dictating cellular structural integrity across living systems.

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE) For questions regarding any of these points, contact directly: Phone: +234 7072 4774 WhatsApp: <https://wa.link/y4sk57>